

CLOUD STORAGE EXPLORER USING AWS S3

Internship Project Report

Intern Details

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Duration: 4 Weeks

Organization: CodTech IT Solutions

Abstract

Cloud Storage Explorer is a desktop-based application developed using Python and Amazon Web Services (AWS). The application allows users to upload, download, delete, refresh, and manage files stored in an Amazon S3 bucket through a graphical user interface.

The project demonstrates cloud storage management using AWS S3 and Python's Boto3 SDK.

Objectives

- Learn AWS S3 cloud storage.**
 - Integrate Python with AWS.**
 - Build a GUI using CustomTkinter.**
 - Perform CRUD operations on cloud storage.**
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Software Requirements

- **Python 3.x**
 - **Visual Studio Code**
 - **Boto3**
 - **CustomTkinter**
 - **Python-dotenv**
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Hardware Requirements

- **Windows 10/11**
 - **Internet Connection**
 - **AWS Account**
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Technologies Used

- **Python**
 - **AWS S3**
 - **Boto3**
 - **CustomTkinter**
 - **dotenv**
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System Architecture

User



Desktop GUI



Python Application



Boto3 SDK



AWS S3 Bucket

Implementation

The application establishes a secure connection with Amazon S3 using AWS credentials stored in a .env file. Users can perform various file management operations through the graphical interface.

Main functionalities include:

- **Upload files**
- **Download files**

- **Delete files**
 - **Refresh file list**
 - **Display bucket contents**
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Results

The application successfully:

- **Connected to AWS S3**
 - **Listed bucket contents**
 - **Uploaded files**
 - **Downloaded files**
 - **Deleted files**
 - **Refreshed bucket contents**
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Advantages

- **User-friendly interface**
 - **Secure AWS integration**
 - **Simple cloud file management**
 - **Cross-platform Python application**
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Applications

- **Personal cloud storage**
 - **File backup systems**
 - **Educational projects**
 - **Cloud computing demonstrations**
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Future Scope

- **Drag-and-drop uploads**
 - **Image preview**
 - **Search functionality**
 - **Multi-bucket support**
 - **Storage analytics dashboard**
 - **User authentication**
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Conclusion

Cloud Storage Explorer successfully demonstrates the integration of Python desktop development with Amazon S3 cloud storage. The project provides hands-on experience with cloud services, GUI development, and AWS SDK integration, making it a valuable learning experience in cloud computing.
